UTSAV CHAUDHARY

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Senior **Data Engineer** with **7+ years** building, deploying, and supporting **batch and near real-time data pipelines** and data platforms in **regulated utility**, **energy**, and **enterprise consulting** environments, supporting **enterprise analytics**, **reporting**, and **AI/ML** initiatives. Strong in **Python**, **PySpark**, and expert-level **SQL** with **ETL/ELT** frameworks, hands-on with **Snowflake** (**Snowpipe**, **Streams**, **Tasks**, **Snowpark**), **change data capture**, **data quality** and reconciliation frameworks, **dimensional modeling**, and **performance and cost optimization** across **AWS** and **Azure**, delivered through **Agile** teams with **CI/CD** and **data governance**.

Technical Skills

Languages & Programming: Python, SQL, PySpark, TypeScript, JavaScript, Bash

Data Engineering: ETL/ELT frameworks, Snowflake (Snowpipe, Streams, Tasks, Dynamic Tables, Snowpark), change data capture, incremental processing, dimensional modeling, data warehousing, data quality and reconciliation, data governance, data preparation and feature engineering

Pipelines, Streaming & APIs: Batch and streaming patterns, Apache Kafka, WebSockets, Python FastAPI, Flask, Pydantic, Pandas, REST APIs, scheduled orchestration

Databases & Storage: Snowflake, PostgreSQL, MongoDB, Redis, Azure SQL, SQL Server, AWS S3, Azure Blob Storage

Cloud, DevOps & CI/CD: AWS (EC2, EKS, Lambda, RDS, S3, VPC, IAM, API Gateway, CloudWatch), Azure (AKS, App Services, Azure SQL, Key Vault, Azure DevOps), Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, Git, environment promotion, automated testing, Agile

Security & Compliance: RBAC, SSL/TLS, OAuth 2.0, NERC CIP, ISO 27001, PCI, OWASP, secure coding, vulnerability assessments, privacy and regulatory requirements

AI/ML Support: Azure OpenAI, Llama, Hugging Face, RAG, embeddings, feature engineering for ML workloads, TensorFlow

Experience

Sr. Data Engineer / Technical Lead, Hybrid, **MidAmerican Energy** (Contract: Midh), TX — Feb 2025 to Present

- Led parallel **data platform** and product workstreams for a regulated electric and natural gas utility, turning business requirements into technical specifications for **data pipelines** across grid telemetry and smart meter (AMI) sources.
- Built **batch and near real-time ingestion** from ERP, customer information system (CIS), and market data feeds into **Snowflake**, with **Snowpipe** for continuous micro-batch loads and **Tasks** for scheduled transformations.
- Implemented **change data capture** and incremental processing with **Streams**, watermark columns, and merge logic, replacing full reloads, cutting pipeline run time by 60%; used **Snowpark** to move Python transformation logic close to data.
- Designed the **data quality** and **reconciliation** framework: row count and checksum validations, schema drift checks, and freshness monitoring so only curated, verified records served analytics and **AI/ML** workloads.
- Modeled **dimensional star schemas** for outage management and AMI metering marts; tuned queries, clustering keys, and warehouse sizing for **performance and cost efficiency**.
- Led **code reviews**, design reviews, and architecture reviews keeping pipelines consistent on **Python**, **SQL**, automated testing, coding standards, and **data governance** aligned to **NERC CIP** compliance reviews.
- Streamed **Kafka** telemetry into Python transformation services on EKS/AKS, and delivered through **Agile** teams with **CI/CD** (**GitHub Actions**, Bitbucket) across globally distributed stakeholders.

Data Engineer, On-Site, **Chevron**, GA — Jan 2024 to Jan 2025

- Built and reviewed **Python ETL** for **drilling reporting** and refinery maintenance data landing in **Azure SQL**, with **automated testing** before **AKS** promotion, helping keep 99.9% pipeline availability behind Application Gateway.
- Delivered **near real-time telemetry ingestion** (Kafka, WebSockets) into Python transformation services and **React.js** operational views, keeping schemas and contracts intact through environment promotion on Azure.
- Wrote and tuned **expert SQL** over large **seismic and operational datasets**, applying dimensional models to reporting marts and lifting query performance by 40%.
- Added **data quality checks** and **reconciliation** between source systems and marts, with **Azure Key Vault** encryption and RBAC on proprietary data, cutting downstream data incidents by 75%.
- Supported **environment promotion** through **Azure DevOps** pipelines, Terraform, and Git workflows, containerizing **FastAPI** data services with **automated testing** and **OWASP**-aligned secure coding.
- Built **pipeline observability** with Application Insights, Azure Monitor, and Grafana, supporting MTTD from 30 minutes to 5 minutes; led root cause analysis and **vulnerability assessments** with Risk and Compliance.

Data Engineer CO-OP, On-Site, **Midh Technologies**, TX — June 2023 to Dec 2023

- Built **Python ETL** and **Pandas** jobs that profiled, cleaned, and loaded curated tables into **PostgreSQL** and **MongoDB** for dashboards, APIs, and **feature engineering** for **Llama** and Hugging Face model fine-tuning.

- Cut manual data preparation time by 50% with reusable Python transformation modules, with **Pydantic** schemas, retries, and validation gating bad records before load.
- Served curated data through **Python FastAPI** and **Flask REST APIs** behind AWS API Gateway / Azure API Management, with IAM authentication, request throttling, and **PostgreSQL** query tuning.
- Developed **React.js** analytics dashboards with TypeScript and D3.js over warehouse tables with granular **RBAC**, translating analyst requirements into production reporting workflows.
- Built production **CI/CD** for pipeline code with GitHub Actions and Azure DevOps: **pytest** unit tests, Docker image builds, security scanning, Terraform, and environment promotion.

Data Engineer, On-Site, **Accenture**, India June 2021 to July 2022

- Built **ELT pipelines** in Python and **Pandas** for enterprise client accounts in a global consulting engagement, loading curated data into **PostgreSQL** behind interactive D3.js and Chart.js reporting that improved data visualization efficiency by 45%.
- Optimized **SQL** queries and **Python FastAPI** data endpoints with Pydantic models, sustaining 250ms average response time behind Nginx with SSL/TLS termination on hybrid on premise deployments.
- Built **Python** scraping/ETL with Selenium, BeautifulSoup, and Pandas across diverse source systems, cutting manual collection time by 60%; added WebSocket streaming so operational dashboards stayed current.
- Implemented self hosted **CI/CD** with GitHub Actions and Jenkins for pipeline and API code, plus Prometheus/Grafana monitoring; applied code reviews and **ISO 27001**-aligned data handling, reducing operational overhead by 60%.

Software Engineer, Data Pipelines, Remote, **VRoKCs**, MO March 2020 to May 2021

- Built **Python** data pipelines for **data collection, annotation, and dataset preparation** feeding **TensorFlow Lite** sign language models, with image upload and webcam frames landing in **Azure Blob Storage**.
- Deployed **Flask** ingestion and prediction APIs on **Azure App Services** with auto scaling and integrated monitoring, and batch preprocessing through **Azure DevOps CI/CD**, reducing deployment cycles by 40%.
- Applied **automated testing** and secure coding on critical path data endpoints; partnered with ML engineers on Python preprocessing and **Docker** for inference services.

Software Developer, Data & Backend, On-Site, **Wonderbiz.in**, India March 2019 to Feb 2020

- Built **Industrial IoT sensor data ingestion** in **Python Flask** and **Node.js**, landing high-frequency **sensor data** and **asset tracking** records into **SQL Server** for the **Angular** analytics dashboard.
- Automated **SQL Server** backups, **sensor data** cleanup, and reporting with Python scripts, and used Git versioning with Jenkins builds for **environment promotion** across Dev, Test, and Production.
- Performed code reviews, unit and integration **automated testing**, and secure coding checks (input validation, auth, logging) before Industry 4.0 releases to customer environments.

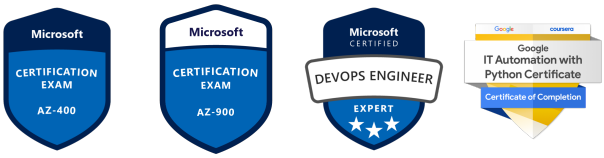
Backend Developer, On-Site, **underDogs Gaming Studios**, India Jan 2018 to Feb 2019

- Built game service **REST APIs** in **Python** and **Node.js** over **SQL** stores, integrating the Quarters API for wallets, currency conversion, and secure transactions across multiple titles.
- Wrote **Python** tooling and JavaScript admin screens for payments, inventory, and shipping, keeping transactional data reliable when requests failed or the network dropped.
- Applied API security, logging, and integration **automated testing**, and stood up release and hosting for game backends with Git source control and health checks.

Education

State University of New York at Binghamton Aug 2022 to Dec 2023
Master of Science in Computer Science 3.4/4.00 GPA

Certifications



Microsoft Certified: DevOps Engineer Expert View Credential
AZ-400: Designing and Implementing Microsoft DevOps Solutions
AZ-900: Microsoft Certified: Azure Fundamentals
Google IT Automation with Python (Specialization) View Credential